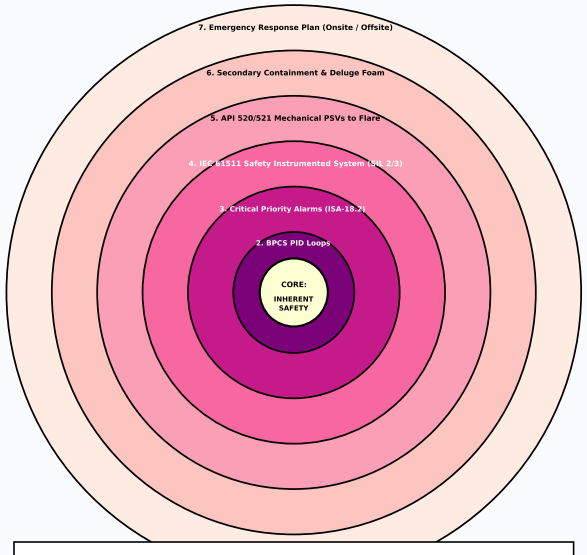
PROCESS SAFETY MANAGEMENT (PSM), IEC 61511 SIS LAYERS & API 521 FLARE RADIATION

LOPA PROTECTION ONION | ESD CAUSE & EFFECT LOGIC | API 521 DEPRESSURING DYNAMICS & THERMAL FLUX

LAYER OF PROTECTION ANALYSIS (LOPA)



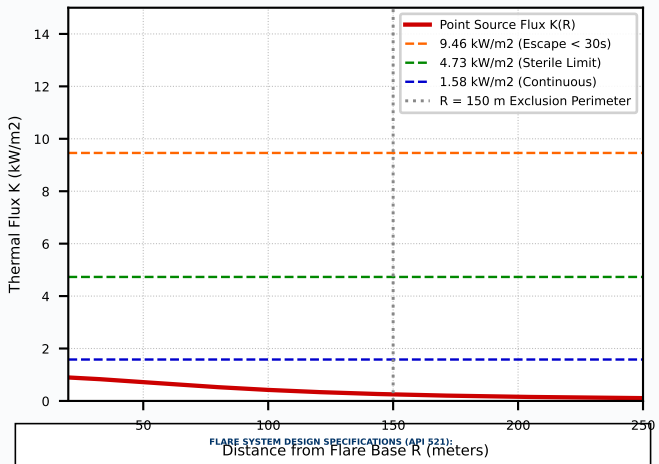
LOPA TARGET RISK MITIGATION:

Unmitigated Event Frequency: 1.0E-01 / yr

Independent Protection Layers (IPL) RRF = 10^4

Mitigated Consequence: < 1.0E-05 / yr (Broadly Acceptable)

API 521 FLARE THERMAL RADIATION PROFILE



FLARE SYSTEM DESIGN SPECIFICATIONS (API 521):

Stack Height: 90 m (Derrick Supported) | Tip Diameter: 1.8 m

Smokeless Flaring Capacity: 25% Design Load with MP Steam Injection

Knockout Drum (V-601): Sized for 300 um Droplet Disengagement (API 521)

Molecular Seal: Purge Gas Consumption < 0.015 m/s to Prevent Air Ingress

EMERGENCY SHUTDOWN (ESD) MATRIX

ESD-101: CDU OVERPRESSURE

Trigger Cause: PT-101 > 3.5 bar(g) (2oo3 Voting)

-> Close Fuel Gas XV-101 (Furnace trip)

-> Trip Feed Pump P-101A/B

-> Open Flare Blowdown XV-102

ESD-201: VDU COIL LOSS OF FLOW

Trigger Cause: FT-201 < 60% Design Flow

-> Trip Vacuum Heater H-201 Burners

-> Inject Emergency Quench Steam

-> Isolate Residue Bottoms Discharge

ESD-301: FCC RISER THERMAL TRIP

Trigger Cause: TT-301 > 560 C (Riser Runaway)

-> Close Regen Catalyst Slide Valve XV-301

-> Inject Lift Steam Dispersion Quench

-> Divert HVGO Feed to Slop System

ESD-401: HCU HP LOOP DEPRESSURING

Trigger Cause: PT-401 > 170 bar OR TT-401 > 425 C

-> Trip Recycle H2 Compressor C-401

-> Open Rapid Depressuring XV-401 to Flare

-> Rate: 155 bar to 10 bar in 15 min (API 521)

SAFETY INSTRUMENTED SYSTEMS (SIS): IEC 61508 / IEC 61511 RELIABILITY & SIL ASSIGNMENT MATRIX

SIF LOOP TAG	INITIATOR / SENSOR	VOTING ARCH.	TARGET SIL	PFD_avg (PROB. FAIL)	SAFE STATE ACTION & EXECUTIVE RESPONSE
SIF-101 (CDU Pressure)	Triple PT Transmitters (PT-101A/B/C)	2oo3 Voting	SIL 2	2.4E-03 (RRF 416)	Cut fuel gas to H-101 within 2.0s; vent column to flare
SIF-102 (CDU Level High)	Dual Displacer Level (LT-101A/B)	1oo2 Voting	SIL 2	3.1E-03 (RRF 322)	Trip reflux pumps to prevent overhead condenser flooding
SIF-201 (VDU Furnace)	Differential Flow FT-201A/B	1oo2 Voting	SIL 3	8.5E-04 (RRF 1176)	Immediate fuel gas trip & emergency coil steam injection
SIF-301 (FCC Slide Valve)	Nuclear Bed Density & Riser Temp	2oo3 Voting	SIL 3	5.2E-04 (RRF 1923)	Slam-shut regenerated catalyst slide valve in < 1.0s
SIF-401 (HCU Depressuring)	High-Pressure Loop PT-401A/B/C	2oo3 Voting	SIL 3	4.1E-04 (RRF 2439)	Open twin fail-open depressuring valves to flare (API 521)
SIF-501 (Coker Switch)	Drum Level Gamma Ray Detectors	2oo3 Voting	SIL 2	1.8E-03 (RRF 555)	Inhibit 4-way switch valve movement during active filling
SIF-601 (Flare Pilot Loss)	Triple Optical / Thermocouple Flame	2oo3 Voting	SIL 2	2.2E-03 (RRF 454)	Auto-spark igniter sequence & emergency fuel gas pilot enrich

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS PLANT DIGITAL TWIN SIMULATION

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PROCESS DYNAMICS, PROTECTION LAYERS & FLARE SAFETY SYSTEM	PROCESS PLANT DIGITAL TWIN & CAD	B.E. CHEMICAL ENGINEERING	DIAGRAMMATIC / ANALYTIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/VP/DWG-06	19 / 10 / 2026

VIRTUAL PETROLEUM REFINERY | 100,000 BPD CRUDE DIET | GAZEBO & ROS 2 DIGITAL TWIN | API / ASME BPVC